Math 101 — Exam 4 Extra Credit

Part 1 (1 point): Complete all 25 provided questions on separate paper. Show handwritten steps. Red answers are for checking only.

Part 2 (2 points): Create and solve 25 similar questions, showing handwritten work. You may ask for help but must submit your own work.

Total extra credit: 3 points.

1. Evaluate $\log_2\left(\frac{1}{32}\right)$.

Answer: -5

2. Evaluate $\log_3(81)$.

Answer: 4

3. Rewrite $4^3 = 64$ in logarithmic form.

Answer: $\log_4(64) = 3$

4. Rewrite $\log_5\left(\frac{1}{25}\right) = -2$ in exponential form.

Answer: $5^{-2} = \frac{1}{25}$

5. Solve using the quadratic formula: $2x^2 + 3x - 2 = 0$.

Answer: $x = \frac{1}{2},\ -2$

6. Solve using the quadratic formula: $3x^2 + 2x - 1 = 0$.

Answer: $x = \frac{1}{3},\ -1$

7. Solve $(x-4)^2 = 7$ using the square root property.

Answer: $x = 4 \pm \sqrt{7}$

8. Solve $(x+3)^2 = 20$ using the square root property.

Answer: $x = -3 \pm 2\sqrt{5}$

9. Solve $9x^2 + 36 = 0$ in complex numbers.

Answer: $x = \pm 2i$

10. Solve $4x^2 + 25 = 0$ in complex numbers.

Answer: $x = \pm\frac{5}{2}i$

11. Find the vertex and axis of $y = -x^2 + 6x - 5$.

Answer: Vertex $(3, 4)$; axis $x = 3$

12. Find the vertex and axis of $y = (x+2)^2 - 3$.

Answer: Vertex $(-2, -3)$; axis $x = -2$

13. For $f(x) = \sqrt{2x+11}$ and $g(x) = x - 3$, find $f(g(x))$.

Answer: $\sqrt{2x+5}$; domain $x \geq -\frac{5}{2}$

14. For $f(x) = 2x + 1$ and $g(x) = x^2 - 4$, find $g(f(x))$.

Answer: $4x^2 + 4x - 3$

15. For $y = \log_2(x+3)$, state the vertical asymptote and one point.

Answer: Asymptote $x = -3$; point $(-2, 0)$

16. For $y = -3^x$, give the horizontal asymptote and y-intercept.

Answer: Asymptote $y = 0$; $(0, -1)$

17. Find the midpoint of $(-4, 7)$ and $(6, -1)$.

Answer: $(1, 3)$

18. Find the distance between $(1, 2)$ and $(4, 6)$.

Answer: 5

19. For $f(x) = 2|x-3| - 4$, state the vertex and opening.

Answer: $(3, -4)$; opens upward

20. Graph $f(x) = \begin{cases} -2, & x < 1 \\ 3, & x \geq 1 \end{cases}$. Mark endpoints.

Answer: Open at $(1, -2)$; closed at $(1, 3)$

21. Find the maximum of $h(t) = -5t^2 + 30t + 10$ and when it occurs.

Answer: 55 at $t = 3$

22. Solve by completing the square: $x^2 + 4x - 1 = 0$.

Answer: $x = -2 \pm \sqrt{5}$

23. Find the value of \$2,000 at 5% interest compounded continuously for 3 years.

Answer: $A = 2000e^{0.15} \approx \$2{,}323.67$

24. Graph $y = -(x-2)^2 + 3$. State the vertex and axis.

Answer: Vertex $(2, 3)$; axis $x = 2$

25. For $y = \log_3(x-2)$, state the domain and vertical asymptote.

Answer: Domain $(2, \infty)$; asymptote $x = 2$